

Folddisco 3.0

Discontinuous motif search across the protein universe. This release adds a sensitivity preset, amino-acid substitution, a confident-hits filter, and new default ranking — measured on 742 M-CSA catalytic sites and the published zinc-finger and serine-peptidase protocols.

branch feature-integration b907df5 **baseline** origin/master 2a756d9

host 20 cores, NVMe, warm cache **indices** 2.x compatible

M-CSA SENSITIVITY @ 1ST FP

0.4961 ↑ from 0.4475

Default search, 250 catalytic sites. 133 queries improve, 42 lose.

QUERY TIME

2.65 s ↓ from 5.35 s

Mean per M-CSA query against the 62,122-entry PDB index.

CONFIDENT HIT LIST

0.76 precision

Up from 0.06 unfiltered; median list 1,804 → 21 structures.

MUTATED QUERY RECOVERY

+0.177 Sens@1FP

One catalytic residue renamed; **:*** wins back most of the loss.

Every number on this page comes from a single rerun of the benchmark suite at commit **b907df5** ; the scripts and raw tables are listed at the end.

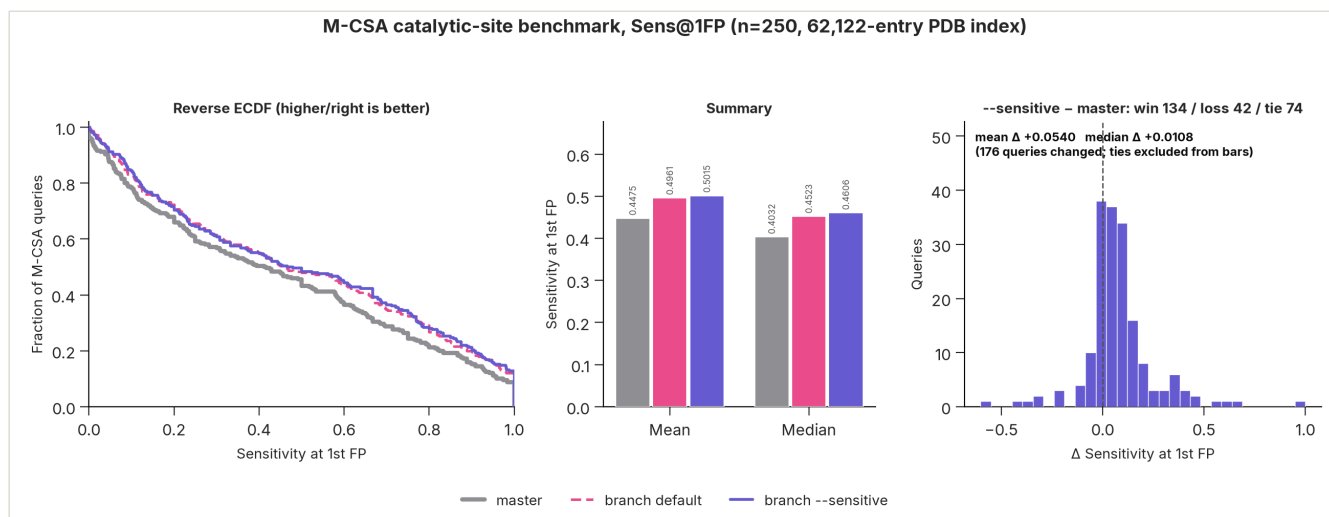
What 3.0 adds

FEATURE	FLAG	WHAT IT DOES
Sensitive search	<code>--sensitive</code>	A residue pair may fall in neighbouring distance/angle bins, several features at once
Amino-acid substitution	<code>--aa-subst</code> , <code>:*</code>	Matches chemically similar residues, scored below exact ones
Confident hits	<code>--confident</code>	Keeps only full, low-RMSD matches instead of every partial one
Index-time expansion	<code>index --expand-radius</code>	Stores the neighbourhood in the index instead of expanding the query
Novelty screening	<code>--novelty-mode</code>	One KNOWN / PARTIAL / NOVEL row per query, with its evidence and the best hit's residues
Multi-character chains	<code>-q AA_250</code> , <code>--chain-sep</code>	mmCIF chain IDs that are not a single letter; 2.x crashed on these files
Deformation metrics	<code>d_rmsd</code>	Superposition-free deviation, as column, sort key and filter
Mapped lookup caches	automatic	53.7 M-entry lookup loads in 17.5 ms instead of 2.98 s

Catalytic-site search

M-CSA: 250 catalytic sites of 3–21 residues searched against a 62,122-entry PDB index, scored against the M-CSA homologues of each site. Sensitivity at the first false positive is a k=1 metric, so paired win/loss counts are given alongside.

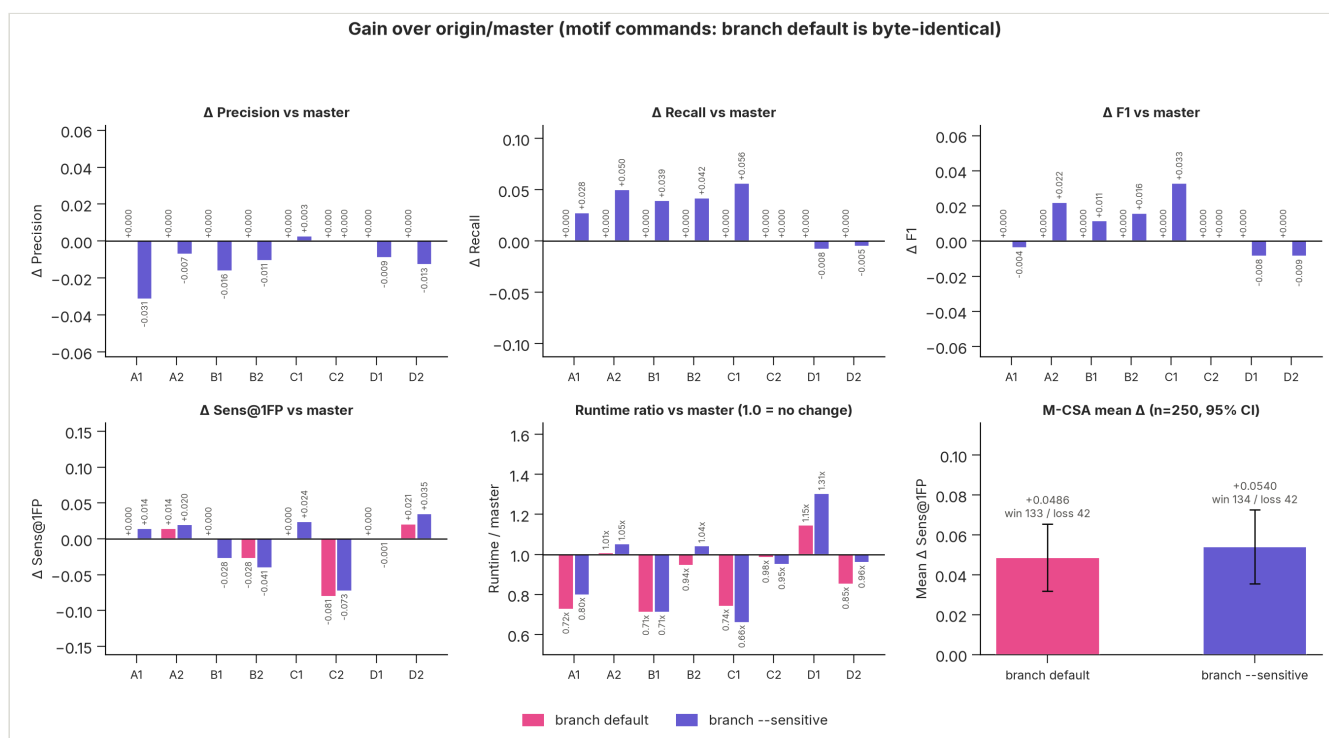
CONFIG	SENS@1FP	Δ MASTER	WIN / LOSS	NO TP	S / QUERY
master (2.x)	0.4475	—	—	10	5.35
3.0 default	0.4961	+0.0486	133 / 42	1	2.65
<code>--sensitive</code>	0.5015	+0.0540	134 / 42	1	3.02
<code>--aa-subst blosum62</code>	0.4764	+0.0289	124 / 56	1	5.69
<code>--aa-subst group</code>	0.4734	+0.0259	120 / 60	1	6.54
<code>--aa-subst size</code>	0.4662	+0.0188	105 / 67	3	6.79
<code>--sensitive --aa-subst</code>	0.4807	+0.0333	120 / 61	2	5.55
Queries that returned no true positive at all fall from 10 to 1.					



M-CSA sensitivity. Reverse ECDF over 250 queries, summary means and the paired difference against master.



Colour follows the configuration in every figure that follows.



Gain over 2.x on the eight motif commands and on M-CSA, with 95% confidence intervals.

Sensitive search: what it does

Folddisco turns each residue pair into a hash by quantising its geometry — two distances and three angles — into bins. Lookup is exact on that hash, so a pair whose distance sits a tenth of an Ångström from a bin edge in the query, and just across it in the target, produces a different hash and the pair is missed. Motifs deform; boundaries do not move with them.

Why a match is missed. Two of the five quantised features are drawn. The default search already covers each single-feature neighbour; a pair that drifts across two boundaries at once needs the diagonal bin, which is what `--sensitive` (`--expand-radius 2`) adds.

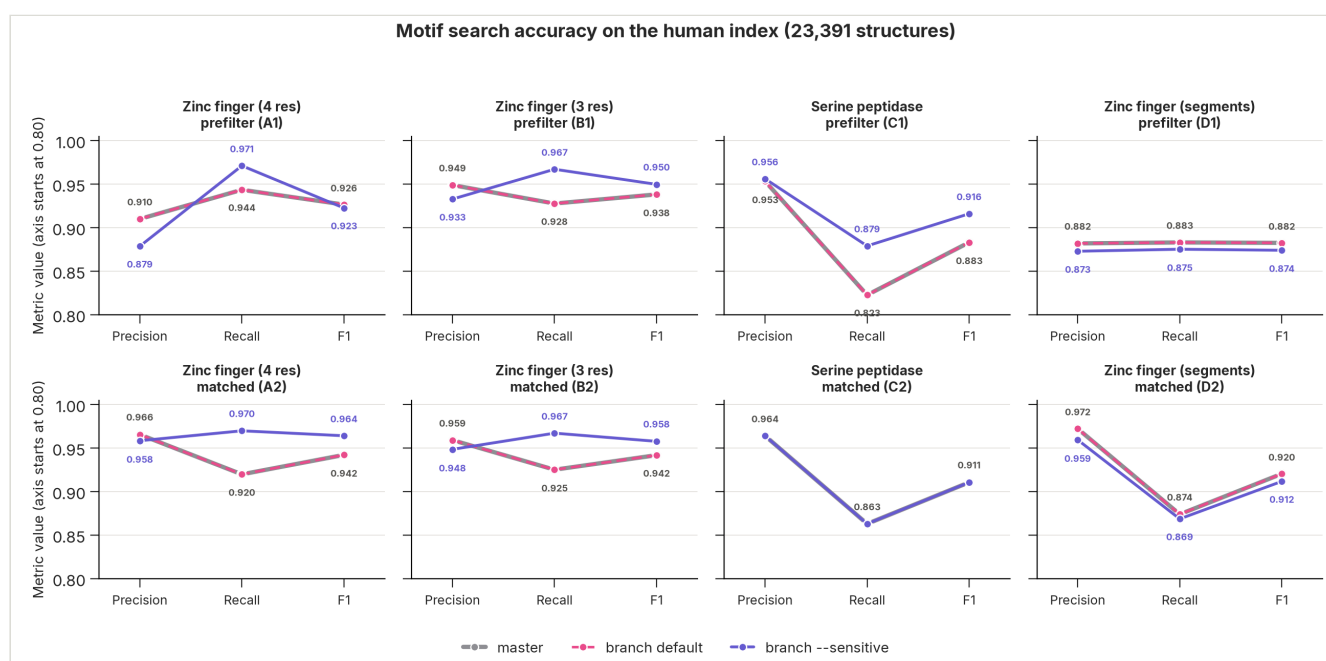
The radius is the number of features allowed to sit in a neighbouring bin simultaneously, not how far a feature may move — `-d` and `-a` set the distance each feature may travel, and wide tolerances are sub-stepped so no bin in between is skipped. Cost on M-CSA: 3.02 s per query against 2.65 s, for +0.0054 sensitivity over the default and +0.054 over 2.x. It is worth pairing with `--max-node`: the extra recall costs precision otherwise.

Motif search on a proteome

Four motifs (3–23 residues) searched against 23,391 human AlphaFold structures under the published protocol: zinc fingers against 761 accessions, the Ser-His-Asp triad against 124 MEROPS S01 accessions. The branch returns the same structures as 2.x on all eight commands; F1 is therefore unchanged, and only the order differs.

COMMAND	F1, MASTER	F1, <code>--SENSITIVE</code>	Δ
Zinc finger, 4 residues, matched	0.9421	0.9641	+0.022
Zinc finger, 3 residues, matched	0.9418	0.9577	+0.016
Ser-His-Asp triad, prefilter	0.8831	0.9160	+0.033
Two-segment zinc, 23 residues, matched	0.9204	0.9117	-0.009

Sensitive search helps short motifs and costs a little on long segment queries.



Motif accuracy per family and mode. Every value sits between 0.82 and 0.98, so the panels are connected dots on an axis starting at 0.80; master and the branch default coincide.

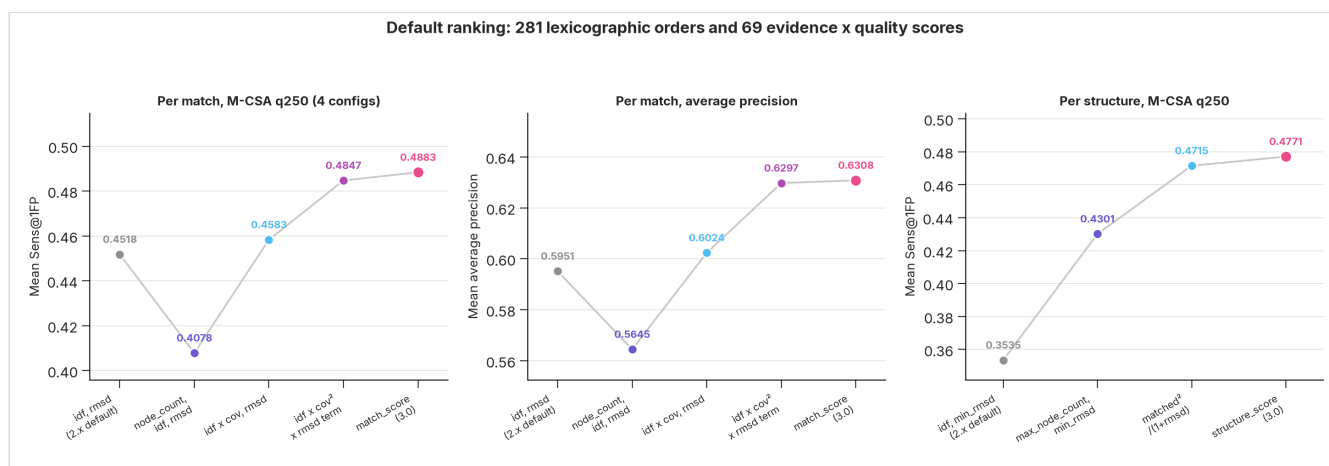
Ranking: how results are ordered

2.x ranked matches by IDF, then RMSD — a heuristic. We searched 281 lexicographic orders over every implemented metric and 69 composite scores of the form evidence $\times$ quality, scoring each on M-CSA and the motif set, then validated the winners on 492 M-CSA queries never used for selection.

PER-MATCH ORDER	SENS@1FP	AVG PRECISION	MOTIF SENS@1FP
<code>idf, rmsd (2.x)</code>	0.4518	0.5951	0.1817
best plain alternative	0.4523	0.5951	0.1837
<code>node_count, idf, rmsd</code>	0.4078	0.5645	0.1958
<code>idf $\times$ coverage, rmsd</code>	0.4583	0.6024	0.2172
<code>match_score = idf $\times$ coverage² $\times$ TM-score</code>	0.4883	0.6308	0.2300

Means over four M-CSA configurations on the 250-query selection set.

On the held-out 492 queries the new order beats 2.x by +0.028 to +0.054 Sens@1FP and +0.034 to +0.050 average precision, every 95% interval excluding zero, at roughly 2.5 wins per loss. Per structure the same shape wins: `structure_score` = $\text{matched}^2 \times \sqrt{\text{idf}} / (1 + \text{RMSD})$, at 0.4863 against 0.4321 for coverage-first and 0.3699 for 2.x.



Ranking selection. The winners are products of evidence and geometric quality, not key orders; dots on a zoomed axis, since the candidates differ by 0.02–0.08.

The motif set disagrees on one metric: with a coverage filter already applied, coverage is constant and `structure_score` reduces to $\sqrt{\text{idf}} / (1 + \text{RMSD})$, which ranks two of four queries worse at the very first false positive (average precision is unchanged at 0.878). For family-level searches with a coverage filter, `--sort-by max_node_count,min_rmsd` restores the coverage-first order.

Confident hits

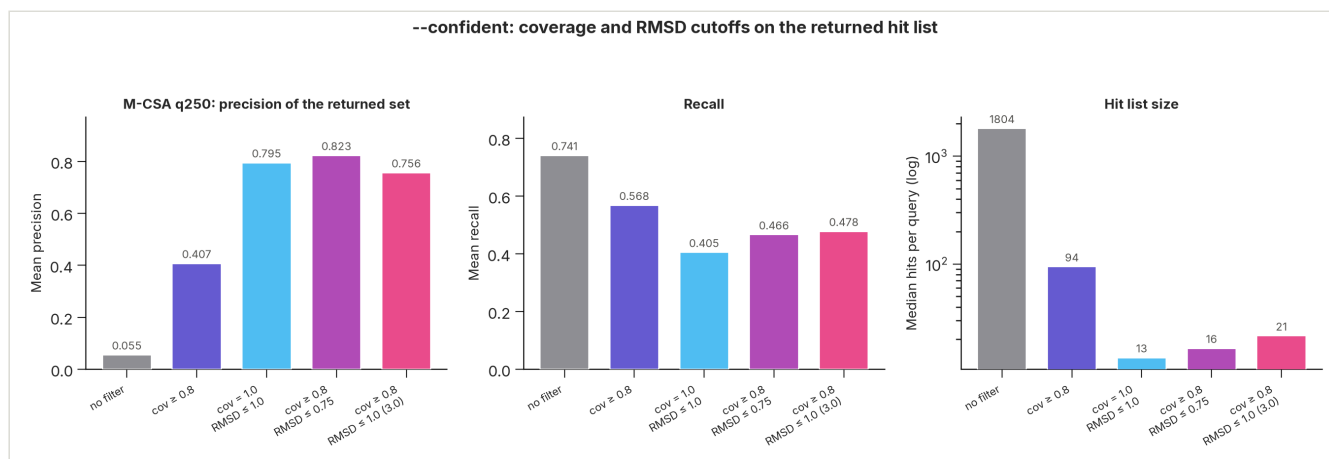
By default a query returns every structure sharing part of the motif — a median of 1,804 structures per M-CSA query, 6% of them correct. `--confident` keeps matches covering at least 80% of the query residues (all of them for a 3–4 residue motif) within 1 Å RMSD; past 12 residues, where matches are assembled from several parts, coverage alone decides.

M-CSA Q250	PRECISION	RECALL	F1	MEDIAN HITS	QUERIES WITH A HIT
default output	0.058	0.751	0.084	6,000	100%
<code>--confident</code>	0.757	0.475	0.508	55	92%
<code>--confident --sensitive</code>	0.750	0.484	0.510	63	92%
Hit counts are output rows; the 55 rows are about 21 distinct structures. Runtime is unchanged.					

On the zinc-finger and serine-peptidase protocols one flag reproduces filters that were tuned per query by hand — `--covered-node 3 --max-node 4 --rmsd 1.0` and friends:

MOTIF (HUMAN PROTEOME)	FILTER	PRECISION	RECALL	F1	HITS
Zinc finger, 4 residues 1G2F F207,F212,F225,F229	none	0.492	0.974	0.654	7,425
	protocol, by hand	0.966	0.920	0.942	746
	<code>--confident</code>	0.961	0.928	0.944	4,304
Zinc finger, 3 residues 1G2F F207,F225,F229	none	0.551	0.972	0.703	7,171
	protocol, by hand	0.959	0.925	0.942	756
	<code>--confident</code>	0.903	0.958	0.930	5,744
Serine peptidase triad 4CHA B57,B102,C195	none	0.245	0.919	0.387	556
	protocol, by hand	0.964	0.863	0.911	113
	<code>--confident</code>	0.964	0.863	0.911	116
Zinc finger, 23-residue segments 1G2F F204-215,F222-232	none	0.038	1.000	0.072	404,416
	protocol, by hand	0.972	0.874	0.920	697
	<code>--confident</code>	0.996	0.356	0.525	407
Answer sets: 761 zinc-finger and 124 MEROPS S01 accessions. Hits are output rows.					

The triad case is exact — same precision, same recall, 116 rows against 113 — and the 4-residue zinc finger beats the hand-tuned filter on F1. On the 23-residue segment query `--confident` is deliberately precision-first: 0.996 precision at 0.356 recall, where the hand-tuned `--max-node 15` reaches 0.972 at 0.874. Lower the bar with `--confident --max-node-ratio 0.65` when recall matters more.



Cutoff selection. Coverage ≥ 0.8 with RMSD ≤ 1 Å was chosen on 250 queries and confirmed on 492 held-out ones.

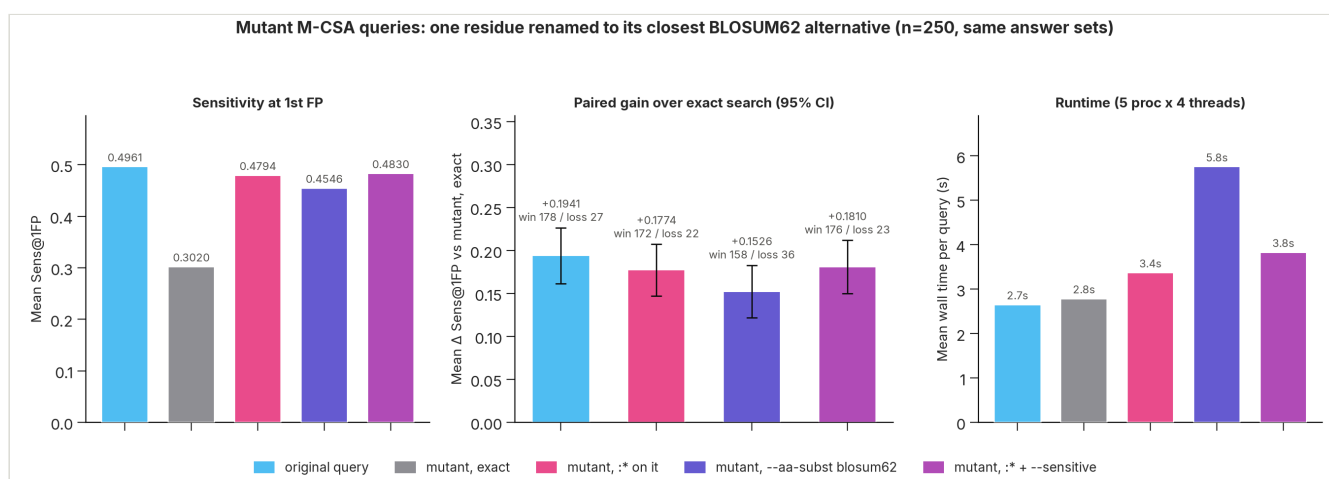
Amino-acid substitution

Substituted residues are scored below exact ones — weight 0.75 per substituted residue, IDF capped at the query's own pair, one substituted hit per residue pair — so exact matches keep the top of the list. Without that weighting, substitution ranked noise from common residues above exact matches and cost 0.11 Sens@1FP.

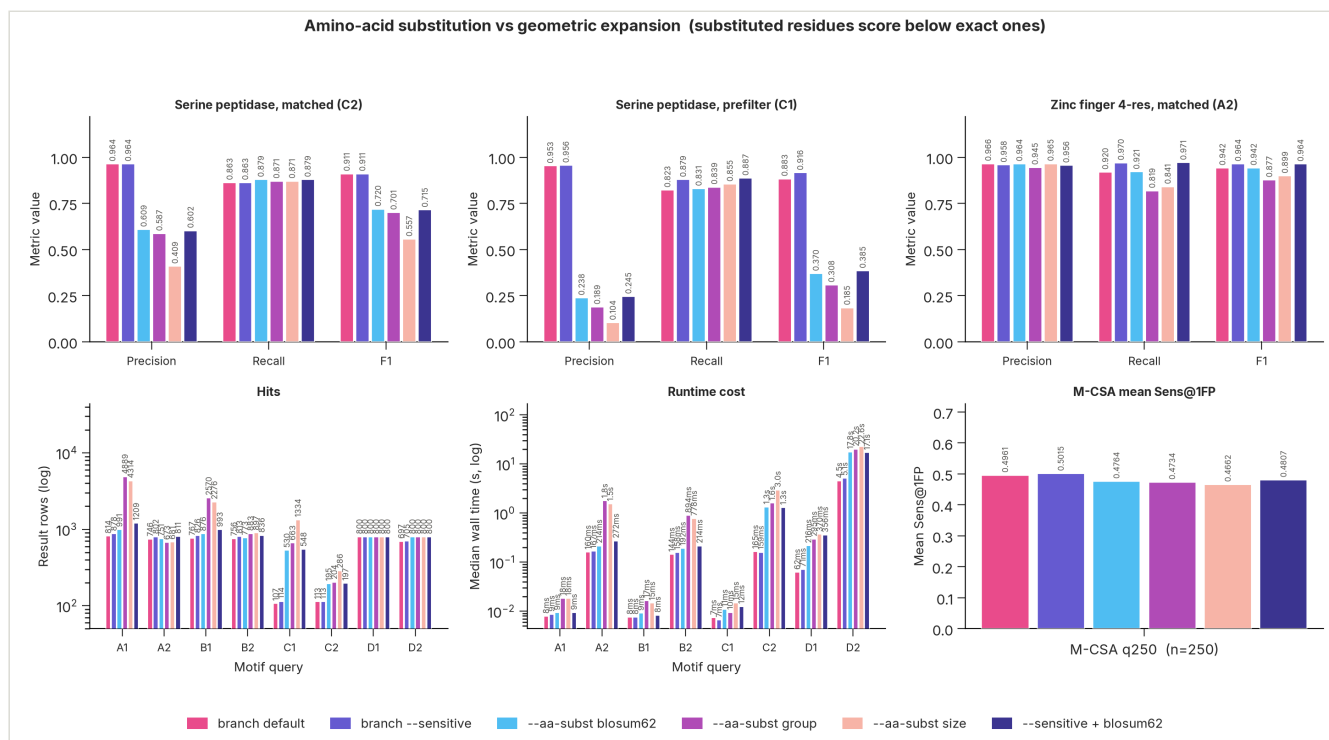
To measure what substitution buys, each M-CSA query had one catalytic residue renamed to its closest BLOSUM62 alternative (His→Tyr, Asp→Glu, ...), leaving the answer sets untouched.

MUTATED QUERY, 250 SITES	SENS@1FP	Δ VS EXACT SEARCH	WIN / LOSS	S / QUERY
original query (reference)	0.4961	—	—	2.65
mutated, exact search	0.3020	—	—	2.78
mutated, <code>:*</code> on that residue	0.4794	+0.177	172 / 22	3.37
mutated, <code>--aa-subst blosum62</code>	0.4546	+0.153	158 / 36	5.76
mutated, <code>:*</code> + <code>--sensitive</code>	0.4830	+0.181	176 / 23	3.83

All intervals exclude zero; renaming one residue costs 0.194 and substitution wins back most of it.



Mutated queries. The gain is paired per query, with 95% confidence intervals.

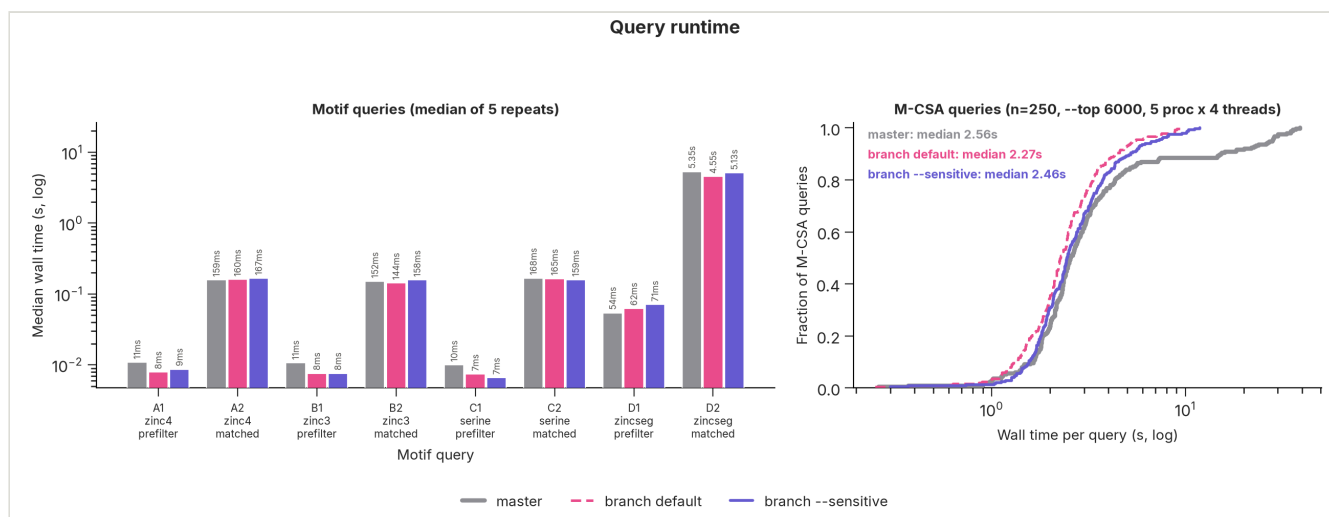


Substitution schemes on the motif protocol and M-CSA. Wider schemes cost precision and time.

On answer sets defined by exact residue identity, substitution is neutral at best — which is what the M-CSA table above shows. It is worth its cost when the residues may differ.

Speed

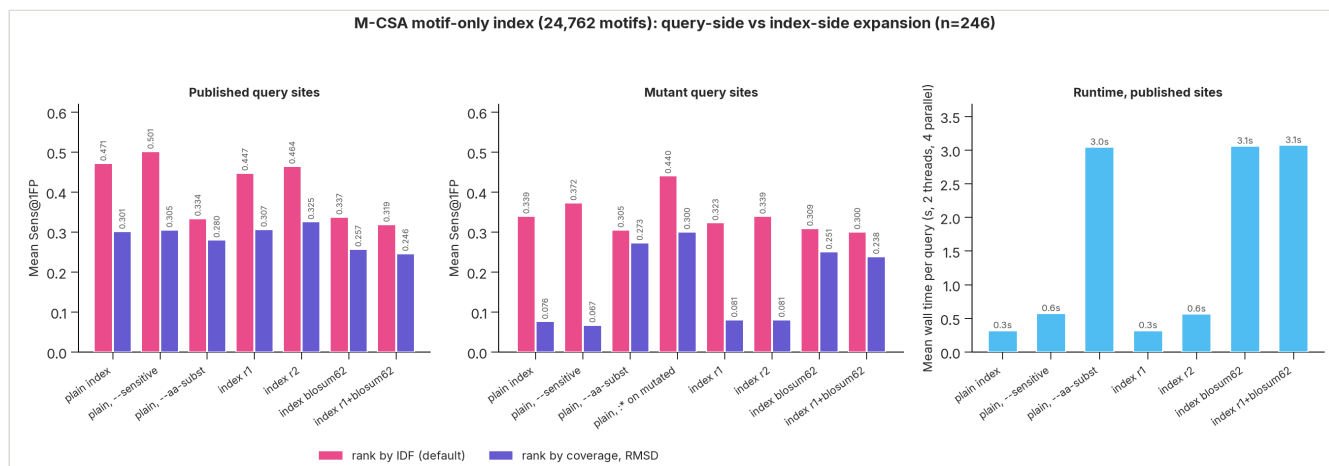
Matching now pre-selects target residues by amino-acid code for large queries, with byte-identical output. Default M-CSA queries run in half the time of 2.x, and sensitive search costs 3.0 s where the first implementation cost 9.6 s.



Runtime. Motif queries (log scale) and the M-CSA distribution across 250 queries.

Index-time expansion

Expansion can also be baked into an index. On a motif-only index built from 24,762 annotated M-CSA sites, index-side expansion is not a substitute for expanding the query: it scores 0.025–0.037 lower, no query returns the same ranking, and the returned motifs overlap only 43–58%. Shifting the query and shifting the targets reach different residue pairs near bin edges. The index also grows 3.5× at radius 1 and 32× with substitution.



Motif-only index. Query-side expansion (left bars) against indices built with expansion.

Multi-character chain IDs

Large mmCIF assemblies label chains **AR**, **AW** or **10**, not **A**. 2.x read the wrong field for these files and crashed on them; 3.0 keeps the author chain ID, up to eight characters, and accepts it in a query. Because **AR119** is ambiguous, multi-character and numeric chains take a separator:

```
$ folddisco query -i complexdir_folddisco -p 4V8S-assembly1.cif.gz \
-q AR_119,AR_364,AR_392 --format-output tid,node_count,match_score,rmsd,matching_residues

4V8S-assembly1.cif.gz 3 21.2189 0.0000 AR_119,AR_364,AR_392
4AYB-assembly1.cif.gz 3 15.4677 0.1150 B119,B364,B392
1TWG-assembly1.cif.gz 2 1.4679 0.4712 _A219,A214
```

The motif — Ile119, Phe364, Ile392 of the two-letter chain **AR** — finds its own assembly at 0 Å and the same three residues on the single-letter chain **B** of a homologous assembly at 0.115 Å, then a partial two-residue match. Each hit is spelled in its own convention: **AR_119** keeps the separator because the chain needs it, **B119** does not, and **_** marks a query residue with no match. **--chain-sep** forces the explicit form everywhere (**A_81,A_128,A_231**), which is what you want when a downstream parser expects one shape.

The same query on 2.x ends in panicked at `src/structure/io/cif.rs`: Chain name should be provided. Leaving the separator out is now a diagnostic rather than a wrong answer: Query 'AR119' names chain 'A' and chain 'R' in 'R119'. A range stays in one chain, and a multi-character or numeric chain needs the separator, as in 'AA_250'.

Default output and index files are unchanged for single-letter chains, so existing pipelines and indices keep working. A related fix: the residue that ends a chain used to be filed under the next chain, so 4CHA **A11** was reported as **B11**; matched-residue labels on multi-chain entries change accordingly.

Also in 3.0

Shorter items, for completeness.

Novelty screening: KNOWN, PARTIAL, NOVEL

`--novelty-mode` prints one verdict row per query instead of a hit list, with the evidence behind it and the residues the best hit matched:

```
$ folddisco query -q designs.txt -i index/mcsa_subset_folddisco --novelty-mode --novelty-rmsd 0.5 --header
```

query_id	verdict	candidates	hits	index_coverage	best_hit
best_coverage	best_rmsd	best_residues			
query/4CHA.pdb	KNOWN	11656	10925	1.0000	mcsa_subset/M0387/pdb4cha.pdb
1.0000	0.0000	B57,B102,C195			
query/1G2F.pdb	KNOWN	2291	2250	1.0000	mcsa_subset/M0035/pdb6zr5.pdb
1.0000	0.4608	D27,D32,D45,D49			
query/2N6N.pdb	PARTIAL	2326	2324	1.0000	mcsa_subset/M0251/pdb6rla.pdb
1.0000	0.6540	B4234,B4298,B4225			
query/1SU6.pdb	NO_HASHES	0	0	0.0000	NA
NA	NA	NA			

KNOWN needs both enough coverage (`--novelty-coverage`, 0.8) and a close enough best hit (`--novelty-rmsd`, 2.0 Å) — the same residues in another arrangement cover everything and are still a different motif. Covered but failing either is **PARTIAL**, nothing covered is **NOVEL**, and a query whose residues are too far apart to hash is **NO_HASHES** rather than a discovery. `best_residues` writes `_` where the hit missed a query residue, so a 2-of-3 match reads `C1145,_,C1128`.

The evidence columns come from the index before filters: `candidates` and `index_coverage` show what was there to match, so a NOVEL verdict caused by strict filters is visible in the row (and warned on `stderr`). Short motifs cover 1.0 against something almost anywhere, so tighten `--novelty-rmsd` when screening 3–4 residue designs — the 2N6N row above is KNOWN at the 2.0 Å default and PARTIAL at 0.5 Å.

Deformation metrics

`drmsd` and `max_dist_deviation` compare internal distances without superposition, which suits hinged motifs; both work as output columns, sort keys and filters. On a synthetic 20° hinge the default search returns a 3-residue mis-assignment at 2.81 Å while `--sensitive` recovers all 8 residues at 1.31 Å.

Lookup caches

Index and Foldcomp lookup tables are memory-mapped caches built on first use, so a large database stops re-parsing its lookup on every query.

TABLE	2.X LOAD	3.0 LOAD	MEMORY
AFDB50 v4 index lookup, 53.7 M entries	2.98 s	17.5 ms	8.55 GB → 2.10 GB
AFDB v6 Foldcomp lookup, 241 M entries	6.14 s	45.8 ms	27.9 GB → 5.66 GB
Caches are regenerable and validated against the source file's size and mtime.			

Fixes and removals

- Substituted residues (`164:H`) passed the index lookup and were then dropped at matching — they now match.
 - A mistyped residue range (`A204-21500000000`) is reported instead of being killed by the out-of-memory reaper; a range may repeat its chain (`F204-F215`).
 - Building without the `foldcomp` feature works again.
 - Removed after measurement: torsion-ENM conformer sampling, the rare-hash IDF filter and `--dist-ratio`, none of which beat the simpler path.
-

Compatibility

- Indices built with 2.x are read unchanged; no rebuild.
 - `--nonrigid` is now `--sensitive`.
 - Hit lists come back in a new order. On the eight motif commands the branch returns exactly the same structures as 2.x; four commands return them reordered.
 - A residue ending a chain is no longer labelled with the next chain's ID, changing matched-residue labels on multi-chain entries.
 - `-d / -a` with several values now use only the widest.
-

Limits

- Sensitivity at the first false positive moves on single accessions; paired counts and average precision are reported with it.
 - M-CSA answers are homologous entries, so a partial match can be correct; the zinc and serine sets are defined by exact residue identity.
 - The coverage-only rule past 12 residues rests on four queries.
 - Index-time expansion was measured on one motif-only index.
 - One machine, one hash type (`PDBTrRosetta`, default binning).
-

Reproduce

```
cd fd-branchbench
REPEATS=5 WARMUP=1 ./scripts/run_all.sh 250      # motif, M-CSA, mutant, --confident, figures
defaults/scripts/collect.sh <binary>              # raw output for ranking and filter selection
defaults/scripts/analyze.py; defaults/scripts/composite2.py; defaults/scripts/plot.py
index_expansion/scripts/run_queries.sh            # motif-only index study
```

Tables: `fd-branchbench/result/` (motif, M-CSA, mutant, confident, answer classes, provenance), `defaults/result/` (ranking and filter grids), `index_expansion/result/`. Measurement detail lives in `folddisco/docs/feature_evaluation.md`, branch state in `docs/UPDATE_REPORT.md`.

Folddisco 3.0 · benchmark rerun at commit b907df5, 18 September 2026 · 20 cores, local NVMe, warm cache.